Arbitrary Convergence Time Control of Unmanned Aerial Manipulator for Active Interaction and Dynamic Tracking

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***Abstract*—This article investigates the problem of aerial active interaction for unmanned aerial manipulator under unknown contact force and lumped disturbances composed of modeling uncertainties and environmental interferences. A dynamic tracking interaction control architecture with arbitrary convergence time is proposed to implement both compliant aerial interaction and stable tracking flight. Particularly, an arbitrary convergence time nonlinear disturbance observer is designed to estimate interaction force in the absence of force sensors. A Takagi–Sugeno–Kang fuzzy estimator with *k*-closest fuzzy rule interpolation is constructed to derive an approximation in representing lumped disturbances. Also, an arbitrary convergence time control strategy is introduced for the quadrotor to implement the rapid response and precise trajectory tracking control, and an admittance controller within arbitrary convergence time for the onboard manipulator is presented to guarantee stable compliant interaction. The stability of the proposed control architecture is analyzed through Lyapunov theory. The feasibility and performance of this novel control architecture are demonstrated through simulations and real-world investigation on tracking and touching mobile robot dynamically.**

***Index Terms*—Aerial active interaction, arbitrary convergence time control, fuzzy estimator, fuzzy rule interpolation (FRI), unmanned aerial manipulator (UAM).**

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I. Introduction

In recent years, unmanned aerial manipulators (UAMs) have attracted lots of attention owing to their advanced maneuverability and superior agility. They have been widely adopted in various complex scenarios and challenge applications for aerial operations, such as agricultural monitoring, object grasping, and industrial infrastructure inspection and maintenance [1]. With the rapid development of the aerial robot technology, the research field of UAM has gradually expanded to support the implementation of active physical interactions with the environment, such as air-ground collaborative operations of multiple robots and human–robot physical interaction [2].

However, maintaining stable physical compliant interaction in dynamic environments is a challenge for UAM in aerial active interaction, specifically regarding the effective compensation of interaction force and the stability for UAM under unknown complex interferences. In the relevant literature, various attempts have been made to estimate the external disturbances due to physically contact with the environment. For instance, an onboard force/torque sensor mounted at the end effector has been exploited to reject external disturbances [3]. Although reliable measurement of contact force could be provided by installing force/torque sensors, the control accuracy is affected by the noise in force/torque sensors. Besides, the mass of the onboard manipulator is generally increased by installing sensors, leading to issues with the center of mass offset. In [4], a force-sensorless contact force control method has been presented to control the force through simultaneous control of position error and pitch angle for unmanned aerial vehicles. It has been noted that nonlinear disturbance observer (NDOB) has been an effective method for rapid estimation and compensation of unknown interferences owing to the fast response and smooth estimation, using joint information of the manipulator to estimate disturbance forces online [5]. An adaptive sliding mode disturbance observer has also been proposed to compensate for the system uncertainties for space manipulator [6], while an NDOB has been adopted to estimate and compensate for uncertainties and external disturbances for UAM [7].

The flight stability of UAM is influenced by lumped disturbances composed of environmental interferences and model uncertainties. To deal with such challenges, estimator construction with known input information emerges as an effective method for the compensation of unknown disturbances. In [8], an adaptive model

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uncertainty estimator has been introduced to provide stable torque input even in the presence of instantaneous disturbances. Particularly, the utilization of fuzzy systems for uncertainty estimation has become a prominent approach in handling nonlinear systems owing to their explicable and comprehensible nature. In [9], a fuzzy neural network estimator has been designed to tune the nonlinear dynamic function vector in fuzzy local models for an n-link robot manipulator. A fuzzy logic system has been utilized to estimate the system uncertainties for a medium-scale unmanned helicopter with six degrees of freedom [10].

Whilst gaining significant popularity, application of fuzzy systems faces the challenges of having to deal with a typically rather large rule base that completely covers the problem domain. This causes a dramatic increase in computational burden as well as a demand for substantial knowledge concerning the application problem. Yet, obtaining a complete knowledge base is extremely difficult, and the construction of the fuzzy estimator relies on the availability of extensive (and often expensive) fuzzy knowledge for the systems designer. In response to these challenges, fuzzy rule interpolation (FRI) has been proposed, which is capable of dynamical generating new fuzzy rules with existing sparse approximate knowledge [11]. As a benefit, feasible estimation and classification with high efficiency may be facilitated under a limited initial fuzzy rule base [12]. For instance, a novel and effective k-closest fuzzy rule interpolation (k-FRI) approach utilizing a small number of closest rules has recently been developed [13], improving the computational efficiency of the interpolative process without assuming full knowledge coverage of the problem domain. From this, the significant potential of k-FRI based on Takagi–Sugeno–Kang (TSK) model [14] has been emerging for high-efficiency lumped disturbance estimation with limited prior knowledge.

For real-world applications of complex operational task, rapid response and precise control for UAM are usually required in trajectory tracking with disturbance compensation. In [15], a rational polynomial tracking control approach has been proposed to stabilize the flight of unmanned aerial vehicle. A model-based adaptive motion control algorithm for an aerial robotic manipulator has been presented to operate in the joint and task spaces [16]. Although stable operation is implemented by such a controller, rapid response capability and error convergence within a finite time are still considered challenging issues. A novel nonsingular terminal sliding mode surface has therefore been introduced to ensure the global convergence of system states to the origin in finite time for n-link robotic manipulators [17]. An adaptive finite-time control scheme with the prescribed performance has been implemented for the specified transient and steady-state performance with the finite-time convergence [18]. A practical finite-time fast adaptive backstepping attitude control method has been realized for aerial manipulators to achieve rapid error convergence and strong immunity of attitude control, based on variable coupling disturbance compensation [19]. In [25], an arbitrary-time sliding mode controller has been developed for n-link robotic manipulator system to ensure exact trajectory tracking within arbitrary time. In [26], an adaptive fuzzy controller has been designed for multiagent systems to globally track the state of the leading robot within predefined time. In [27], a state feedback-based control scheme has been proposed to realize prescribed-time tracking control. These attempts effectively contribute to the implementation of trajectory tracking for UAM, and the design of control strategy with finite or arbitrary convergence time is one of the approaches to explore for improving control performances of UAM in complex operational tasks.

Inspired by the above observations, a dynamic tracking interaction control architecture within arbitrary convergence time is herein proposed to implement compliant interaction and stable tracking flight. To fully demonstrate the advantages of the proposed method, the main contributions of this article are outlined as follows.

1) Precise estimation with rapid response for the unknown interaction force at arbitrary convergence time is implemented, resulting in an arbitrary convergence time nonlinear disturbance observer (ACTNDO). The estimation error convergence time is arbitrarily finite, and the upper bound of interaction force is not strictly required, including the first-order derivative of force and the inertial matrix bounds of the manipulator dynamics.
2) Robust estimation of the complex lumped disturbances can be achieved using the proposed TSK fuzzy estimator (TSKFE) with k-FRI. Specifically, estimation accuracy is enhanced by generating new rules with sparse rule base, and TSKFE is suitable for estimating nonlinear terms with unknown bounds.
3) Stable and precise control for the UAM in question is guaranteed through the dynamic tracking interaction control architecture. As an advantage, the convergence time of the proposed control strategy is arbitrary in contrast to asymptotic convergence and finite/fixed time convergence controllers, demonstrating fast response and excellent control performances.

The remainder of this article is organized as follows. Section II introduces the dynamics of the UAM under consideration. Section III presents the dynamic tracking interaction control architecture. Simulations and real-world experiments reveal the feasibility of the proposed control architecture in Sections IV and V, respectively. Finally, Section VI concludes this article and discusses future work.

II. Dynamics Modeling

This section presents the dynamics of the UAM system under investigation, encompassing the dynamics of the quadrotor and the onboard manipulator. The inertia frame and the quadrotor body-fixed frame are denoted as $\Omega_W : \{O_W, X_W, Y_W, Z_W\}$ and $\Omega_B : \{O_B, X_B, Y_B, Z_B\}$, respectively. The position and velocity of the quadrotor in the inertial frame are denoted by $\boldsymbol{p}_\mathrm{b} = [x, y, z]^\mathrm{T}$ and $\boldsymbol{\nu}_\mathrm{b}$, respectively. The attitude of the quadrotor in the inertial frame is denoted by $\Phi_\mathrm{b} = [\phi, \theta, \psi]^\mathrm{T}$ and $\boldsymbol{\omega}_\mathrm{b}$ is the angular velocity.

The trajectory of quadrotor is denoted as $\boldsymbol{\chi} = [\boldsymbol{p}_\mathrm{b}, \Phi_\mathrm{b}]^\mathrm{T}$. Considering the impact of external interferences during interaction with environment and lumped disturbances, the dynamics model of the quadrotor can be established by Newton–Euler theory [20]:

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$$\boldsymbol{M}_{\mathrm{u}}\ddot{\boldsymbol{\chi}}+\boldsymbol{C}_{\mathrm{u}}\dot{\boldsymbol{\chi}}+\boldsymbol{G}_{\mathrm{u}}+\boldsymbol{\Delta}_{\mathrm{u}}(\boldsymbol{\chi},\dot{\boldsymbol{\chi}})=\boldsymbol{\tau}_{\mathrm{u}}+\boldsymbol{J}_{\mathrm{E}}\boldsymbol{\tau}_{\mathrm{ext}} \tag{1}$$

where $\boldsymbol{M}_{\mathrm{u}}=\mathrm{diag}(m_{\mathrm{u}}\boldsymbol{R}_{\mathrm{b}}^{-1},\boldsymbol{J}_{\mathrm{b}}\boldsymbol{T}^{-1})$ is the inertia matrix; $\boldsymbol{C}_{\mathrm{u}}=\mathrm{diag}(\boldsymbol{O}_{3\times3},-\boldsymbol{J}_{\mathrm{b}}\boldsymbol{T}^{-1}\dot{\boldsymbol{T}}\boldsymbol{T}^{-1})$ and $\boldsymbol{O}_{3\times3}$ is a zero matrix; $\boldsymbol{G}_{\mathrm{u}}=\mathrm{diag}(-m_{\mathrm{u}}\boldsymbol{R}_{\mathrm{b}}^{-1}\boldsymbol{G}_{\mathrm{p}},\boldsymbol{T}^{-1}\dot{\Phi}_{\mathrm{b}}\times\boldsymbol{J}_{\mathrm{b}}\boldsymbol{T}^{-1}\dot{\Phi}_{\mathrm{b}})$ is the gravity vector; $\boldsymbol{J}_{\mathrm{b}}=\mathrm{diag}(I_x,I_y,I_z)$ is the inertia matrix and $\boldsymbol{T}$ is the transformation matrix; m_{u} is the mass of quadrotor and $\boldsymbol{R}_{\mathrm{B}}$ is the rotation matrix from body-fixed frame to inertia frame; $\boldsymbol{\Delta}_{\mathrm{u}}(\boldsymbol{\chi},\dot{\boldsymbol{\chi}})$ represents lumped disturbances composed of modeling uncertainties and environment interferences; $\boldsymbol{\tau}_{\mathrm{u}}\in\mathbb{R}^{6\times1}$ is the control input of quadrotor; $\boldsymbol{\tau}_{\mathrm{ext}}=[\tau_{\mathrm{ext}1},\ldots,\tau_{\mathrm{ext}n}]^{\mathrm{T}}$ represents the external interaction force exerted on the joints of onboard manipulator; and $\boldsymbol{J}_{\mathrm{E}}=\boldsymbol{R}_{\mathrm{E}}\boldsymbol{R}_{\mathrm{F}}\boldsymbol{J}_{\mathrm{m}}^{-\mathrm{T}}$ is the force transformation matrix, $\boldsymbol{R}_{\mathrm{F}}=[\boldsymbol{I}_{3\times3},\boldsymbol{O}_{3\times3}]^{\mathrm{T}}$, $\boldsymbol{J}_{\mathrm{m}}$ is the Jacobian matrix of n-joint manipulator, and $\boldsymbol{R}_{\mathrm{E}}$ is denoted as follows [21]:

$$\boldsymbol{R}_{\mathrm{E}}=\begin{bmatrix}\boldsymbol{I}_{3\times3} & \boldsymbol{O}_{3\times3}\\ \boldsymbol{T}^{\mathrm{T}}(\Phi_{\mathrm{b}})\boldsymbol{S}(\boldsymbol{p}_{\mathrm{e}}^{\mathrm{b}}) & \boldsymbol{T}^{\mathrm{T}}(\Phi_{\mathrm{b}})\end{bmatrix} \tag{2}$$

where $\boldsymbol{S}(\cdot)$ is the skew-symmetric matrix operator and $\boldsymbol{p}_{\mathrm{e}}^{\mathrm{b}}$ represents the position of the end-effector for onboard manipulator relative to Ω_B of quadrotor.

Particularly, the control input of quadrotor $\boldsymbol{\tau}_{\mathrm{u}}$ is expressed as $\boldsymbol{\tau}_{\mathrm{u}}=[\upsilon_1,\upsilon_2,\upsilon_3,u_4,u_5,u_6]^{\mathrm{T}}$. υ_1, υ_2, and υ_3 are the virtual inputs of the position controller, and u_4, u_5, and u_6 are the attitude control inputs. For underactuated quadrotor, the desired roll and pitch angles are derived through the attitude decoupling module based on position controller and desired yaw angle. The attitude decoupling module is shown as follows [28]:

$$\begin{cases} f=\sqrt{\upsilon_1^2+\upsilon_2^2+\upsilon_3^2}\\ \phi_{\mathrm{d}}=\sin^{-1}[\dfrac{1}{f}(\upsilon_1\sin\psi_{\mathrm{d}}-\upsilon_2\cos\psi_{\mathrm{d}})]\\ \theta_{\mathrm{d}}=\tan^{-1}[\dfrac{1}{\upsilon_3}(\upsilon_1\cos\psi_{\mathrm{d}}+\upsilon_2\sin\psi_{\mathrm{d}})]\end{cases} \tag{3}$$

where f is the thrust of quadrotor. ϕ_{d}, θ_{d}, and ψ_{d} are the desired roll, pitch, and yaw angles, respectively.

For an n-joint onboard manipulator, $\boldsymbol{q}=[q_1,\ldots,q_n]^{\mathrm{T}}$ is denoted as the joint angle and $\boldsymbol{q}_{\mathrm{d}}$ is the desired joint angle. Then, the dynamics can be represented as

$$\boldsymbol{M}_{\mathrm{m}}(\boldsymbol{q})\ddot{\boldsymbol{q}}+\boldsymbol{C}_{\mathrm{m}}(\dot{\boldsymbol{q}},\boldsymbol{q})\dot{\boldsymbol{q}}+\boldsymbol{G}_{\mathrm{m}}(\boldsymbol{q})=\boldsymbol{\tau}_{\mathrm{m}}+\boldsymbol{\tau}_{\mathrm{ext}} \tag{4}$$

where $\boldsymbol{M}_{\mathrm{m}}\in\mathbb{R}^{n\times n}$, $\boldsymbol{C}_{\mathrm{m}}\in\mathbb{R}^{n\times n}$, and $\boldsymbol{G}_{\mathrm{m}}\in\mathbb{R}^{n\times1}$ are the positive inertia matrix, centrifugal, and Coriolis matrix, and gravity matrix, respectively. The control input is represented as $\boldsymbol{\tau}_{\mathrm{m}}\in\mathbb{R}^{n\times1}$. For the matrices $\boldsymbol{M}_{\mathrm{m}}(\boldsymbol{q})$ and $\boldsymbol{C}_{\mathrm{m}}(\dot{\boldsymbol{q}},\boldsymbol{q})$, some useful properties are presented as follows [25]:

Property 1: The matrix $\boldsymbol{M}_{\mathrm{m}}(\boldsymbol{q})$ is positive-definite.

Property 2: The matrix $\dot{\boldsymbol{M}}_{\mathrm{m}}(\boldsymbol{q})-2\boldsymbol{C}_{\mathrm{m}}(\dot{\boldsymbol{q}},\boldsymbol{q})$ is a skew-symmetric matrix.

III. Controller Design

In this section, the control architecture comprising ACTNDO, TSKFE, and arbitrary convergence time controller is designed to implement active interaction and dynamic tracking, as shown in Fig. 1. The stability analyses are also provided.

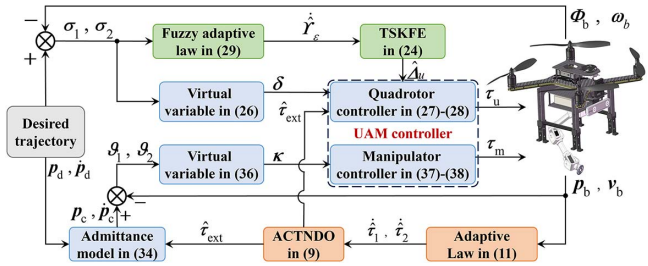


Fig. 1. Structure of proposed dynamic tracking interaction control architecture within arbitrary convergence time for UAM systems.

A. Design of ACTNDO

An ACTNDO is herein designed for the UAM to estimate the interaction force under dynamic environment.

Assumption 1: $\boldsymbol{\tau}_{\mathrm{ext}}$ and the first derivative are assumed to be bounded but unknown. Then, $\boldsymbol{\tau}_1\geq\boldsymbol{\tau}_{\mathrm{ext}}$, $\boldsymbol{\tau}_2\geq\dot{\boldsymbol{\tau}}_{\mathrm{ext}}$ are satisfied with the unknown upper bound $\boldsymbol{\tau}_1$ and $\boldsymbol{\tau}_2$.

Lemma 1: Suppose that there exists a real-valued continuously differentiable function V and a real number $\rho\geq1$. If the conditions $V(0)=0$, $V(x)>0,\forall x\neq0$ are satisfied and the first derivative of V satisfies as follows:

$$\dot{V}\leq\frac{-\rho(\mathrm{e}^V-1)}{\mathrm{e}^V(t_{\mathrm{e}}-t)},\quad\forall t\in[t_{\mathrm{i}},t_{\mathrm{e}}). \tag{5}$$

Then, the equilibrium point $x=0$ is arbitrary time stable [22], where t_{i} is the initial time and t_{e} is the end time.

According to the dynamics of manipulator of (4), the variable Θ is defined as $\Theta=\boldsymbol{M}_{\mathrm{m}}(\boldsymbol{q})\dot{\boldsymbol{q}}$. Introduce the auxiliary error variable $\boldsymbol{s}_{\mathrm{o}}=\Theta-\boldsymbol{\zeta}$ with the variable $\boldsymbol{\zeta}$ designed to satisfy the following constraint:

$$\dot{\boldsymbol{\zeta}}-\boldsymbol{C}_{\mathrm{m}}^{\mathrm{T}}\dot{\boldsymbol{q}}+\boldsymbol{G}_{\mathrm{m}}=\boldsymbol{\tau}_{\mathrm{m}}+\hat{\boldsymbol{\tau}}_{\mathrm{ext}}+\boldsymbol{\nu} \tag{6}$$

where $\boldsymbol{\nu}=-\hat{\boldsymbol{\tau}}_{\mathrm{ext}}+\hat{\boldsymbol{\tau}}_1+\varpi_{\mathrm{o}1}$. $\hat{\boldsymbol{\tau}}_{\mathrm{ext}}$ and $\hat{\boldsymbol{\tau}}_1$ are the estimation of $\boldsymbol{\tau}_{\mathrm{ext}}$ and $\boldsymbol{\tau}_1$, respectively. For the finite time interval $[t_{\mathrm{s}},t_{\mathrm{s}}+\Delta t_{\mathrm{o}1})$ with t_{s} being the initial time of interaction and $\Delta t_{\mathrm{o}1}>0$ denoting the convergence time, then $\varpi_{\mathrm{o}1}$ is designed as

$$\varpi_{\mathrm{o}1}=\begin{cases}\dfrac{\eta_{\mathrm{o}1}(\mathrm{e}^{s_{\mathrm{o}}}-1)}{\mathrm{e}^{s_{\mathrm{o}}}(t_{\mathrm{s}}+\Delta t_{\mathrm{o}1}-t)}, & t_{\mathrm{s}}\leq t<t_{\mathrm{s}}+\Delta t_{\mathrm{o}1}\\ 0, & \text{otherwise}\end{cases} \tag{7}$$

where $\mathrm{e}^{(\cdot)}$ represents the exponential function and $\eta_{\mathrm{o}1}\geq1$.

By substituting the variable Θ and (6), the derivative of $\boldsymbol{s}_{\mathrm{o}}$ is obtained such that

$$\dot{\boldsymbol{s}}_{\mathrm{o}}=\tilde{\boldsymbol{\tau}}_{\mathrm{ext}}-\boldsymbol{\nu} \tag{8}$$

where $\tilde{\boldsymbol{\tau}}_{\mathrm{ext}}=\boldsymbol{\tau}_{\mathrm{ext}}-\hat{\boldsymbol{\tau}}_{\mathrm{ext}}$ is the estimation error of $\boldsymbol{\tau}_{\mathrm{ext}}$.

The ACTNDO is thus designed as follows:

$$\begin{cases}\hat{\boldsymbol{\tau}}_{\mathrm{ext}}=\boldsymbol{\varrho}+\lambda\Theta\\ \dot{\boldsymbol{\varrho}}=-\lambda\left(\boldsymbol{C}_{\mathrm{m}}^{\mathrm{T}}\dot{\boldsymbol{q}}-\boldsymbol{G}_{\mathrm{m}}+\boldsymbol{\tau}_{\mathrm{m}}+\hat{\boldsymbol{\tau}}_{\mathrm{ext}}\right)-\varpi_{\mathrm{o}2}+\hat{\boldsymbol{\tau}}_2\end{cases} \tag{9}$$

where $\lambda>0$ is a positive constant and $\hat{\boldsymbol{\tau}}_2$ is the estimation of $\boldsymbol{\tau}_2$. For the finite time interval $I_o\in[t_{\mathrm{s}},t_{\mathrm{oe}})$, $t_{\mathrm{oe}}=t_{\mathrm{s}}+\Delta t_{\mathrm{o}1}+\Delta t_{\mathrm{o}2}$ is the convergence end time for ACTNDO. Particularly, $\varpi_{\mathrm{o}2}$ is

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designed with $\Delta t_{o2} > 0$ and $\eta_{o2} \geq 1$, such that

$$\varpi_{o2} = \begin{cases} \dfrac{\eta_{o2}(e^{v}-1)}{e^{v}(t_{oe}-t)}, & t_s \leq t < t_{oe} \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

The adaptive law of $\hat{\tau}_1$ and $\hat{\tau}_2$ are designed as

$$\dot{\hat{\tau}}_1 = \Gamma_{o1} s_o, \ \dot{\hat{\tau}}_2 = \Gamma_{o2} v \tag{11}$$

where Γ_{o1} and Γ_{o2} are the positive constant diagonal matrix.

Theorem 1: At arbitrary convergence time $\Delta T_o = \Delta t_{o1} + \Delta t_{o2}$, both the estimation error $\tilde{\tau}_{ext}$ of interaction force and the auxiliary error variable s_o under ACTNDO as expressed by (9) converge to 0, namely, $\tilde{\tau}_{ext} \to 0$ and $s_o \to 0$.

Proof: Define the error $\tilde{\tau}_1 = \tau_1 - \hat{\tau}_1$ and $\tilde{\tau}_2 = \tau_2 - \hat{\tau}_2$. Then, the derivatives are expressed as $\dot{\tilde{\tau}}_1 = -\dot{\hat{\tau}}_1$ and $\dot{\tilde{\tau}}_2 = -\dot{\hat{\tau}}_2$.

Step 1: Consider the following Lyapunov function $V_{o1} = s_o^T s_o + \tilde{\tau}_1^T \Gamma_{o1}^{-1} \tilde{\tau}_1$, where $s_o^T s_o \geq 0$ and $\tilde{\tau}_1^T \Gamma_{o1}^{-1} \tilde{\tau}_1 \geq 0$ are satisfied with positive diagonal matrix Γ_{o1}. Consequently, the inequality $\sqrt{V_{o1}} \geq \sqrt{s_o^T s_o} = \|s_o\|$ can be derived.

Then, the first derivative of V_{o1} can be expressed as

$$\dot{V}_{o1} \leq 2 s_o^T \tilde{\tau}_1 - 2 s_o^T \varpi_{o1} - 2 \tilde{\tau}_1^T \Gamma_{o1}^{-1} \dot{\hat{\tau}}_1. \tag{12}$$

By substituting (7) and (11) into (12), it can be obtained

$$\begin{aligned} \dot{V}_{o1} &\leq -\frac{2\eta_{o1} s_o^T (e^{s_o}-1)}{e^{s_o}(t_s + \Delta t_{o1} - t)} \leq -\frac{2\eta_{o1}\|s_o\|(e^{\|s_o\|}-1)}{e^{\|s_o\|}(t_s + \Delta t_{o1} - t)} \\ &\leq -\frac{2\eta_{o1}\sqrt{V_{o1}}(e^{\sqrt{V_{o1}}}-1)}{e^{\sqrt{V_{o1}}}(t_s + \Delta t_{o1} - t)}. \end{aligned} \tag{13}$$

Denote $\xi_{o1} = \sqrt{V_{o1}}$ and substitute the derivative of ξ_{o1} into (13), it follows that

$$\dot{\xi}_{o1} \leq -\frac{\eta_{o1}(e^{\xi_{o1}}-1)}{e^{\xi_{o1}}(t_s + \Delta t_{o1} - t)} \tag{14}$$

where $\xi_{o1} = 0$ at some time $t_s \leq t < t_s + \Delta t_{o1}$ owing to Lemma 1. The auxiliary error variable s_o converges to 0 at arbitrary convergence time Δt_{o1}. Thereafter, $s_o = \dot{s}_o = 0$ is satisfied all the time and $\tilde{\tau}_{ext} = v$ can be obtained by (8).

Step 2: When $s_o = 0$ is reached after the arbitrary convergence time Δt_{o1}, another Lyapunov function can be constructed with the positive constant diagonal matrix Γ_{o2}, as follows:

$$V_{o2} = \tilde{\tau}_{ext}^T \tilde{\tau}_{ext} + \tilde{\tau}_2^T \Gamma_{o2}^{-1} \tilde{\tau}_2. \tag{15}$$

From (9), it can be obtained that $\dot{\tilde{\tau}}_{ext} = -\lambda \tilde{\tau}_{ext} + \varpi_{o2} - \hat{\tau}_2 + \dot{\tau}_{ext}$. Then, the time derivative of V_{o2} is derived as

$$\dot{V}_{o2} \leq 2\tilde{\tau}_{ext}^T(-\lambda \tilde{\tau}_{ext} + \varpi_{o2} - \hat{\tau}_2 + \tau_2) - 2\tilde{\tau}_2^T \Gamma_{o2}^{-1} \dot{\hat{\tau}}_2. \tag{16}$$

By substituting (10) and (11) into (16), $\dot{V}_{o2}$ is derived as follows:

$$\begin{aligned} \dot{V}_{o2} &\leq -2\lambda \tilde{\tau}_{ext}^T \tilde{\tau}_{ext} - \frac{2\eta_{o2}\tilde{\tau}_{ext}^T(e^{v}-1)}{e^{v}(t_{oe}-t)} \\ &\leq -\frac{2\eta_{o2}\|\tilde{\tau}_{ext}\|(e^{\|\tilde{\tau}_{ext}\|}-1)}{e^{\|\tilde{\tau}_{ext}\|}(t_{oe}-t)} \\ &\leq -\frac{2\eta_{o2}\sqrt{V_{o2}}(e^{\sqrt{V_{o2}}}-1)}{e^{\sqrt{V_{o2}}}(t_{oe}-t)}. \end{aligned} \tag{17}$$

Denote $\xi_{o2} = \sqrt{V_{o2}}$ and the inequality is obtained such that

$$\dot{\xi}_{o2} \leq -\frac{\eta_{o2}(e^{\xi_{o2}}-1)}{e^{\xi_{o2}}(t_{oe}-t)} \tag{18}$$

where $\xi_{o2} = 0$ has been reached at some time $t_s + \Delta t_{o1} \leq t < t_{oe}$ owing to Lemma 1. Consequently, the estimation error $\tilde{\tau}_{ext}$ of ACTNDO converges to 0 at arbitrary convergence time $\Delta T_o = \Delta t_{o1} + \Delta t_{o2}$. The proof is completed. ■

B. Fuzzy Estimator With k-FRI

A TSKFE is herein constructed to estimate lumped disturbances for quadrotor. In particular, the k-closest fuzzy rules interpolation is introduced to enhance the estimation robustness and overall performance of the fuzzy estimator with limited prior knowledge.

Without losing generality, the TSK model with antecedent input variables $\{b_1, ..., b_v\}$ is defined by a set of rules of the following form:

$$\begin{aligned} R_i: \ &\text{if } b_1 \text{ is } A_1^i, ..., b_v \text{ is } A_v^i \\ &\text{then } f_i(b_1, ..., b_v) = a_0^i + a_1^i b_1 + ... + a_v^i b_v \end{aligned} \tag{19}$$

where $a_0^i, a_1^i, ..., a_v^i$ are the parameters specifying the polynomials of a rule's consequent, and $A_1^i, ..., A_v^i$ are the fuzzy sets corresponding to the antecedent input variables.

For the present application problem, the input variables of TSKFE are denoted as $x_1 = \max(|\chi_d - \chi|)$ and $x_2 = \max(|\dot{\chi}_d - \dot{\chi}|)$, where $\max(\cdot)$ represents the maximum value function and χ_d is the desired trajectory. That is, in (19), $v = 2$ with $b_1 = x_1$ and $b_2 = x_2$. Let $\mu_{max} \in [0, 1]$ denotes the judgment threshold for rule firing. An input is considered to match a rule if both $\mu_{max} \leq \max(\mu_{A_1})$ and $\mu_{max} \leq \max(\mu_{A_2})$ are satisfied, where μ_{A_1} and μ_{A_2} are the membership values for inputs. Conversely, a new fuzzy rule is generated through k-FRI for an unmatched input by the known fuzzy rules.

As with the existing FRI literature, for computational simplicity, triangular fuzzy membership function is utilized to define each antecedent fuzzy set, such that $A_r = (a_{r1}, a_{r2}, a_{r3})$ [12]. Then, for a component of the observed input, say O_{x_j}, the similarity metric is denoted as $S_j(A_r, O_{x_j}) = [s_1^j, ..., s_{y_j}^j]$ and is introduced as follows [13]:

$$S_j(A_r, O_{x_j}) = \frac{1}{1 + d_j}, \ d_j = \frac{\sum_{i=1}^{3} |a_{ri} - x_j|}{3}, \ (j = 1, 2) \tag{20}$$

where d_j is the Euclidean distance. Given the above similarity measure, the k-closest known fuzzy rules from the rule base are determined by d_j and the interpolation weights for each of such rules R_i are obtained as $w_i = \min(S_1, S_2)$, $i = 1, ..., k$. Subsequently, the newly interpolated consequent of the unmatched input is constructed as $f_{new}(x_1, x_2) = \alpha_0 + \alpha_1 x_1 + \alpha_2 x_2$, where α_0, α_1, and α_2 are the new consequent parameters obtained by the interpolation weights, as follows:

$$\alpha_0 = \frac{\sum_{i=1}^{k} w_i a_{i0}}{\sum_{i=1}^{k} w_i}, \ \alpha_1 = \frac{\sum_{i=1}^{k} w_i a_{i1}}{\sum_{i=1}^{k} w_i}, \ \alpha_2 = \frac{\sum_{i=1}^{k} w_i a_{i2}}{\sum_{i=1}^{k} w_i}. \tag{21}$$

Following the afore-described procedure, a new fuzzy rule is generated. Then, the TSKFE with k-FRI is constructed based on

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$y_1 \times y_2$ initial fuzzy rules, as follows:

$$\begin{aligned}\Delta_{\mathrm{u}}(\varepsilon) &= \boldsymbol{X}\boldsymbol{\Upsilon}_{\varepsilon}\boldsymbol{\zeta}(\boldsymbol{x}_{\mathrm{f}}) \\ &= \frac{\sum_{l_i=1}^{y_1\times y_2} W_{l_i} f_{l_i}(x_1,x_2) + \sum_{l_j=1}^{\rho} W_{l_j} f_{l_j}(x_1,x_2)}{\sum_{l_i=1}^{y_1\times y_2} W_{l_i} + \sum_{l_j=1}^{\rho} W_{l_j}}\end{aligned} \tag{22}$$

where $\Delta_{\mathrm{u}}(\varepsilon)$ is the ε th element of Δ_{u} with $\varepsilon = 1,2,...,6$; ρ is the fuzzy rules generated by k-FRI; $\boldsymbol{x}_{\mathrm{f}} = [x_1,x_2]^{\mathrm{T}}$, $\boldsymbol{X} = [x_1,x_2,1]^{\mathrm{T}}$; $f_{(l_i,l_j)}(x_1,x_2) = a_1x_1 + a_2x_2 + a_0$ is the consequent polynomial; $W_{l_i,l_j} = \boldsymbol{\mu}_{A_1} \cdot \boldsymbol{\mu}_{A_2}$ represents the fuzzy rule strength; the value range of the interpolation parameter is $1 \le k \le y_1 \times y_2$; and $\boldsymbol{\zeta}(\boldsymbol{x}_{\mathrm{f}})$ is the underlying fuzzy membership function. The expression of the polynomial parameter matrix $\boldsymbol{\Upsilon}_{\varepsilon} \in \mathbb{R}^{3\times(y_1\times y_2+\rho)}$ is shown as follows:

$$\boldsymbol{\Upsilon}_{\varepsilon} = \begin{bmatrix} a_{11} & \cdots & a_{(y_1\times y_2)1} & \cdots & a_{\rho 1} \\ a_{12} & \cdots & a_{(y_1\times y_2)2} & \cdots & a_{\rho 2} \\ a_{10} & \cdots & a_{(y_1\times y_2)0} & \cdots & a_{\rho 0} \end{bmatrix}. \tag{23}$$

The estimation of lumped disturbances Δ_{u} for quadrotor is expressed as

$$\hat{\Delta}_{\mathrm{u}} = [\boldsymbol{X}\hat{\boldsymbol{\Upsilon}}_1\boldsymbol{\zeta}(\boldsymbol{x}_{\mathrm{f}}), \boldsymbol{X}\hat{\boldsymbol{\Upsilon}}_2\boldsymbol{\zeta}(\boldsymbol{x}_{\mathrm{f}}), ..., \boldsymbol{X}\hat{\boldsymbol{\Upsilon}}_{\varepsilon}\boldsymbol{\zeta}(\boldsymbol{x}_{\mathrm{f}})]^{\mathrm{T}} \tag{24}$$

where $\varepsilon = 1,2,...,6$. $\hat{\boldsymbol{\Upsilon}}_{\varepsilon}$ is the estimation of the polynomial parameter matrix $\boldsymbol{\Upsilon}_{\varepsilon}$ and $\tilde{\boldsymbol{\Upsilon}}_{\varepsilon} = \boldsymbol{\Upsilon}_{\varepsilon} - \hat{\boldsymbol{\Upsilon}}_{\varepsilon}$ is the estimation error.

C. Arbitrary Convergence Time Controller for Quadrotor

An arbitrary convergence time controller is proposed for quadrotor with the previously obtained ACTNDO and TSKFE. Notably, the zero-error convergence of the estimated interaction force $\tilde{\boldsymbol{\tau}}_{\mathrm{ext}}$ constitutes the essential precondition for implementing the stability of the proposed controller, and the arbitrary convergence time is satisfied as $\Delta T_{\mathrm{o}} < \Delta T_{\mathrm{u}}$ and $\Delta T_{\mathrm{o}} < \Delta T_{\mathrm{m}}$.

In particular, the state error of the quadrotor is obtained from (1), resulting in the following:

$$\begin{cases} \dot{\boldsymbol{\sigma}}_1 = \boldsymbol{\sigma}_2 \\ \dot{\boldsymbol{\sigma}}_2 = \ddot{\boldsymbol{\chi}}_{\mathrm{d}} + \boldsymbol{M}_{\mathrm{u}}^{-1}(\boldsymbol{C}_{\mathrm{u}}\dot{\boldsymbol{\chi}} + \boldsymbol{G}_{\mathrm{u}} + \Delta_{\mathrm{u}}) - \boldsymbol{M}_{\mathrm{u}}^{-1}(\boldsymbol{\tau}_{\mathrm{u}} + \boldsymbol{J}_{\mathrm{E}}\boldsymbol{\tau}_{\mathrm{ext}}) \end{cases} \tag{25}$$

where $\boldsymbol{\sigma}_1 = \boldsymbol{\chi}_{\mathrm{d}} - \boldsymbol{\chi}$ is the tracking error and $\dot{\boldsymbol{\sigma}}_2 = \dot{\boldsymbol{\chi}}_{\mathrm{d}} - \dot{\boldsymbol{\chi}}$.

The virtual variable $\boldsymbol{\delta}$ is designed as

$$\boldsymbol{\delta} = \begin{cases} -\dfrac{\eta_{\mathrm{u}1}(\mathrm{e}^{\boldsymbol{\sigma}_1} - 1)}{\mathrm{e}^{\boldsymbol{\sigma}_1}(t_{\mathrm{u}} - t)}, & t_{\mathrm{s}} \le t < t_{\mathrm{u}} \\ 0, & \text{otherwise} \end{cases} \tag{26}$$

where $t_{\mathrm{u}} = t_{\mathrm{s}} + \Delta T_{\mathrm{u}}$ and $\Delta T_{\mathrm{u}} > \Delta T_{\mathrm{o}}$ is the arbitrary convergence time of the quadrotor.

From ACTNDO as specified in (9) and TSKFE in (24), the arbitrary convergence time controller for quadrotor is designed as

$$\boldsymbol{\tau}_{\mathrm{u}} = \boldsymbol{C}_{\mathrm{u}}\dot{\boldsymbol{\chi}} + \boldsymbol{G}_{\mathrm{u}} + \boldsymbol{M}_{\mathrm{u}}(\ddot{\boldsymbol{\chi}}_{\mathrm{d}} + \boldsymbol{\sigma}_1) + \boldsymbol{M}_{\mathrm{u}}\boldsymbol{\Xi}_{\mathrm{u}} - \boldsymbol{J}_{\mathrm{E}}\hat{\boldsymbol{\tau}}_{\mathrm{ext}} + \hat{\Delta}_{\mathrm{u}} \tag{27}$$

$$\boldsymbol{\Xi}_{\mathrm{u}} = \begin{cases} \dfrac{\eta_{\mathrm{u}1}\mathrm{e}^{-\boldsymbol{\sigma}_1}}{t_{\mathrm{u}} - t}\boldsymbol{\sigma}_2 + \dfrac{\eta_{\mathrm{u}1}(1 - \mathrm{e}^{-\boldsymbol{\sigma}_1})}{(t_{\mathrm{u}} - t)^2} + \dfrac{\eta_{\mathrm{u}2}(\mathrm{e}^{\boldsymbol{\varsigma}} - 1)}{\mathrm{e}^{\boldsymbol{\varsigma}}(t_{\mathrm{u}} - t)}, & t_{\mathrm{s}} \le t < t_{\mathrm{u}} \\ 0, & \text{otherwise} \end{cases} \tag{28}$$

where $\boldsymbol{\varsigma} = \boldsymbol{\sigma}_2 - \boldsymbol{\delta}$ and $\eta_{\mathrm{u}1} \ge 1$, $\eta_{\mathrm{u}2} \ge 1$ are satisfied. The adaptive update law of the fuzzy matrix is proposed as follows:

$$\dot{\hat{\boldsymbol{\Upsilon}}}_{\varepsilon} = c_{\varepsilon}(\boldsymbol{\varsigma}^{\mathrm{T}}\boldsymbol{M}_{\mathrm{u}}^{-1})_{\varepsilon}\boldsymbol{X}^{\mathrm{T}}\boldsymbol{\zeta}^{\mathrm{T}}(\boldsymbol{x}_{\mathrm{f}}), \quad (\varepsilon = 1,2,...,6) \tag{29}$$

where $(\boldsymbol{\varsigma}^{\mathrm{T}}\boldsymbol{M}_{\mathrm{u}}^{-1})_{\varepsilon}$ is the εth element of $\boldsymbol{\varsigma}^{\mathrm{T}}\boldsymbol{M}_{\mathrm{u}}^{-1}$ and c_{ε} is the positive constant number.

Theorem 2: At any convergence time ΔT_{u}, the virtual variable error $\boldsymbol{\varsigma}$ and the estimation errors $\tilde{\boldsymbol{\Upsilon}}_{\varepsilon}$ of the TSKFE all converge to 0 under the control laws (27)–(29) with the disturbance observer described as per (9). The tracking error $\boldsymbol{\sigma}_1$ also converges to 0 with the convergence time and the stability is guaranteed for the quadrotor.

Proof: The first derivative of $\boldsymbol{\varsigma}$ can be derived from (25) and (26), as follows:

$$\dot{\boldsymbol{\varsigma}} = \dot{\boldsymbol{\sigma}}_2 + \frac{\eta_{\mathrm{u}1}\mathrm{e}^{-\boldsymbol{\sigma}_1}\boldsymbol{\sigma}_2}{t_{\mathrm{u}} - t} + \frac{\eta_{\mathrm{u}1}(1 - \mathrm{e}^{-\boldsymbol{\sigma}_1})}{(t_{\mathrm{u}} - t)^2}. \tag{30}$$

Consider the Lyapunov function as $V_{\mathrm{u}} = \boldsymbol{\sigma}_1^{\mathrm{T}}\boldsymbol{\sigma}_1 + \boldsymbol{\varsigma}^{\mathrm{T}}\boldsymbol{\varsigma} + c_{\varepsilon}\sum_{\varepsilon=1}^{6}\mathrm{tr}(\tilde{\boldsymbol{\Upsilon}}_{\varepsilon}^{\mathrm{T}}\tilde{\boldsymbol{\Upsilon}}_{\varepsilon})$. Then, the first derivative of V_{u} can be obtained such that

$$\dot{V}_{\mathrm{u}} = 2\boldsymbol{\sigma}_1^{\mathrm{T}}(\boldsymbol{\varsigma} + \boldsymbol{\delta}) + 2\boldsymbol{\varsigma}^{\mathrm{T}}\dot{\boldsymbol{\varsigma}} - 2c_{\varepsilon}\sum_{\varepsilon=1}^{6}\mathrm{tr}(\tilde{\boldsymbol{\Upsilon}}_{\varepsilon}^{\mathrm{T}}\dot{\hat{\boldsymbol{\Upsilon}}}_{\varepsilon}). \tag{31}$$

By substituting (26)–(29) into (31), the following can be obtained after the estimation error $\tilde{\boldsymbol{\tau}}_{\mathrm{ext}}$ converges to 0 at the arbitrary convergence time ΔT_{o} with $\Delta T_{\mathrm{o}} < \Delta T_{\mathrm{u}}$:

$$\dot{V}_{\mathrm{u}} \le -\frac{2(\eta_{\mathrm{u}1} + \eta_{\mathrm{u}2})\sqrt{V_{\mathrm{u}}}(\mathrm{e}^{\sqrt{V_{\mathrm{u}}}} - 1)}{\mathrm{e}^{\sqrt{V_{\mathrm{u}}}}(t_{\mathrm{u}} - t)}. \tag{32}$$

Denote $\xi_{\mathrm{u}} = \sqrt{V_{\mathrm{u}}}$ and the inequality becomes

$$\dot{\xi}_{\mathrm{u}} \le -\frac{(\eta_{\mathrm{u}1} + \eta_{\mathrm{u}2})(\mathrm{e}^{\xi_{\mathrm{u}}} - 1)}{\mathrm{e}^{\xi_{\mathrm{u}}}(t_{\mathrm{u}} - t)} \tag{33}$$

where $\xi_{\mathrm{u}} = 0$ has been reached at $t_{\mathrm{s}} \le t < t_{\mathrm{u}}$ according to Lemma 1. Consequently, the errors $\boldsymbol{\sigma}_1$, $\boldsymbol{\varsigma}$, and $\tilde{\boldsymbol{\Upsilon}}_{\varepsilon}$ converge to 0 with the arbitrary convergence time ΔT_{u}. ■

D. Admittance Controller With Arbitrary Convergence Time

For effective compliant interaction, the admittance model of the onboard manipulator in (4) is introduced as follows:

$$\boldsymbol{H}_{\mathrm{d}}(\ddot{\boldsymbol{q}}_{\mathrm{c}} - \ddot{\boldsymbol{q}}_{\mathrm{d}}) + \boldsymbol{D}_{\mathrm{d}}(\dot{\boldsymbol{q}}_{\mathrm{c}} - \dot{\boldsymbol{q}}_{\mathrm{d}}) + \boldsymbol{K}_{\mathrm{d}}(\boldsymbol{q}_{\mathrm{c}} - \boldsymbol{q}_{\mathrm{d}}) = \hat{\boldsymbol{\tau}}_{\mathrm{ext}} \tag{34}$$

where $\boldsymbol{q}_{\mathrm{c}}$ is the admittance desired trajectory obtained by $\boldsymbol{q}_{\mathrm{d}}$ and the estimated interaction force $\hat{\boldsymbol{\tau}}_{\mathrm{ext}}$; and $\boldsymbol{H}_{\mathrm{d}}$, $\boldsymbol{D}_{\mathrm{d}}$, and $\boldsymbol{K}_{\mathrm{d}}$ are the mass, damping, and stiffness matrices, respectively.

The tracking error is defined as $\boldsymbol{\vartheta}_1 = \boldsymbol{q}_{\mathrm{c}} - \boldsymbol{q}$. The state errors of onboard manipulator can be derived by (4), as follows:

$$\begin{cases} \dot{\boldsymbol{\vartheta}}_1 = \boldsymbol{\vartheta}_2 \\ \dot{\boldsymbol{\vartheta}}_2 = \ddot{\boldsymbol{q}}_{\mathrm{c}} + \boldsymbol{M}_{\mathrm{m}}^{-1}(\boldsymbol{C}_{\mathrm{m}}\dot{\boldsymbol{q}} + \boldsymbol{G}_{\mathrm{m}}) - \boldsymbol{M}_{\mathrm{m}}^{-1}\boldsymbol{\tau}_{\mathrm{m}} - \boldsymbol{M}_{\mathrm{m}}^{-1}\boldsymbol{\tau}_{\mathrm{ext}} \end{cases} \tag{35}$$

where $\boldsymbol{\vartheta}_2 = \dot{\boldsymbol{q}}_{\mathrm{c}} - \dot{\boldsymbol{q}}$ is the derivative of the tracking error $\boldsymbol{\vartheta}_1$.

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The virtual variable $\boldsymbol{\kappa}$ is designed such that

$$\boldsymbol{\kappa}=\begin{cases}-\dfrac{\eta_{\mathrm{m1}}(\mathrm{e}^{\vartheta_1}-1)}{\mathrm{e}^{\vartheta_1}(t_{\mathrm{m}}-t)}, & t_{\mathrm{s}}\le t<t_{\mathrm{m}}\\ 0, & \text{otherwise}\end{cases} \tag{36}$$

where $t_{\mathrm{m}}=t_{\mathrm{s}}+\Delta T_{\mathrm{m}}$ and $\Delta T_{\mathrm{m}}>\Delta T_{\mathrm{o}}$ is the arbitrary convergence time of the onboard manipulator.

The arbitrary convergence time admittance controller with ACTNDO in (9) is designed as

$$\boldsymbol{\tau}_{\mathrm{m}}=\boldsymbol{C}_{\mathrm{m}}\dot{\boldsymbol{q}}+\boldsymbol{G}_{\mathrm{m}}+\boldsymbol{M}_{\mathrm{m}}(\ddot{\boldsymbol{q}}_{\mathrm{c}}+\boldsymbol{\vartheta}_1)-\hat{\boldsymbol{\tau}}_{\mathrm{ext}}+\boldsymbol{M}_{\mathrm{m}}\boldsymbol{\Xi}_{\mathrm{m}} \tag{37}$$

$$\boldsymbol{\Xi}_{\mathrm{m}}=\begin{cases}\eta_{\mathrm{m1}}\left[\dfrac{\mathrm{e}^{-\vartheta_1}\boldsymbol{\vartheta}_2}{t_{\mathrm{m}}-t}+\dfrac{(1-\mathrm{e}^{-\vartheta_1})}{(t_{\mathrm{m}}-t)^2}\right]+\dfrac{\eta_{\mathrm{m2}}(\mathrm{e}^{\gamma}-1)}{\mathrm{e}^{\gamma}(t_{\mathrm{m}}-t)}, & t_{\mathrm{s}}\le t<t_{\mathrm{m}}\\ 0, & \text{otherwise}\end{cases} \tag{38}$$

where $\boldsymbol{\gamma}=\boldsymbol{\vartheta}_2-\boldsymbol{\kappa}$ and $\eta_{\mathrm{m1}}\ge 1$, $\eta_{\mathrm{m2}}\ge 1$.

Theorem 3: At any convergence time ΔT_{m}, the trajectory tracking error $\boldsymbol{\vartheta}_1$ and the virtual variable error $\boldsymbol{\gamma}$ all converge to 0 under the admittance controller as specified by (37) and (38) with the previously obtained estimation $\hat{\boldsymbol{\tau}}_{\mathrm{ext}}$ as per (9). The effective compliant interaction with exact trajectory tracking for the onboard manipulator is guaranteed.

Proof: Consider the following candidate Lyapunov function $V_{\mathrm{m}}=\boldsymbol{\vartheta}_1^{\mathrm{T}}\boldsymbol{\vartheta}_1+\boldsymbol{\gamma}^{\mathrm{T}}\boldsymbol{\gamma}$. The first derivative of V_{m} is derived

$$\dot{V}_{\mathrm{m}}=2\boldsymbol{\vartheta}_1^{\mathrm{T}}(\boldsymbol{\gamma}+\boldsymbol{\kappa})+2\boldsymbol{\gamma}^{\mathrm{T}}\dot{\boldsymbol{\gamma}} \tag{39}$$

where $\dot{\boldsymbol{\gamma}}$ is obtained by (35) and (36), such that

$$\dot{\boldsymbol{\gamma}}=\dot{\boldsymbol{\vartheta}}_2+\eta_{\mathrm{m1}}\left[\frac{\mathrm{e}^{-\vartheta_1}\boldsymbol{\vartheta}_2}{t_{\mathrm{m}}-t}+\frac{(1-\mathrm{e}^{-\vartheta_1})}{(t_{\mathrm{m}}-t)^2}\right]. \tag{40}$$

Then, by substituting (35)–(38) and (40) into (39), it can be obtained after the estimation error $\tilde{\boldsymbol{\tau}}_{\mathrm{ext}}$ converges to 0 at the arbitrary convergence time ΔT_{o} with $\Delta T_{\mathrm{o}}<\Delta T_{\mathrm{m}}$, as follows:

$$\begin{aligned}\dot{V}_{\mathrm{m}}&=-2\boldsymbol{\vartheta}_1^{\mathrm{T}}\frac{\eta_{\mathrm{m1}}(\mathrm{e}^{\vartheta_1}-1)}{\mathrm{e}^{\vartheta_1}(t_{\mathrm{m}}-t)}-2\boldsymbol{\gamma}^{\mathrm{T}}\frac{\eta_{\mathrm{m2}}(\mathrm{e}^{\gamma}-1)}{\mathrm{e}^{\gamma}(t_{\mathrm{m}}-t)}-2\boldsymbol{\gamma}^{\mathrm{T}}\boldsymbol{M}_{\mathrm{m}}^{-1}\tilde{\boldsymbol{\tau}}_{\mathrm{ext}}\\ &\le-\frac{2(\eta_{\mathrm{m1}}+\eta_{\mathrm{m2}})\sqrt{V_{\mathrm{m}}}(\mathrm{e}^{\sqrt{V_{\mathrm{m}}}}-1)}{\mathrm{e}^{\sqrt{V_{\mathrm{m}}}}(t_{\mathrm{m}}-t)}.\end{aligned} \tag{41}$$

Denote $\xi_{\mathrm{m}}=\sqrt{V_{\mathrm{m}}}$ and the inequality is derived as follows:

$$\dot{\xi}_{\mathrm{m}}\le-\frac{(\eta_{\mathrm{m1}}+\eta_{\mathrm{m2}})(\mathrm{e}^{\xi_{\mathrm{m}}}-1)}{\mathrm{e}^{\xi_{\mathrm{m}}}(t_{\mathrm{m}}-t)} \tag{42}$$

where $\xi_{\mathrm{m}}=0$ has been reached at the time interval $t_{\mathrm{s}}\le t<t_{\mathrm{m}}$ owing to Lemma 1. Consequently, the errors $\boldsymbol{\vartheta}_1$ and $\boldsymbol{\gamma}$ all converge to 0 with the arbitrary convergence time ΔT_{m}. ■

IV. Simulations and Results Analysis

The effectiveness of the present control architecture is herein validated with comparative simulations, addressing the trajectory tracking performance of the implemented UAM, the estimated results of interaction force, and lumped disturbances.

A. Simulation Setup

A two-joint onboard manipulator is taken for simulations. The simulation spans from 0th to 60th seconds, with the quadrotor continuously subjected to lumped disturbances throughout this period. Particularly, the time for interaction force with the dynamic target is set to start at the 10th second and end at the 50th second, the duration of the interaction continues for 40 s. The dynamics model parameters of the UAM system are set as $m_{\mathrm{u}}=1.5$ kg, $\boldsymbol{J}_{\mathrm{b}}=\mathrm{diag}(0.04,0.05,0.06)$ kg $\cdot$ m^2 and $g=9.8\,\mathrm{m/s^2}$. The length of the two rods for the onboard manipulator is set as $l_{\mathrm{m1}}=0.2$ m and $l_{\mathrm{m2}}=0.2$ m. The sparse fuzzy rule base and input fuzzy quantity space are both precompiled, with triangular membership functions chosen as membership functions. The parameters of ACTNDO are set as $\lambda=12$, $\eta_{\mathrm{o1}}=1.25$, $\eta_{\mathrm{o2}}=1.25$. The parameters of TSKFE are set as $k=3$, $\mu_{\max}=0$, $A^i_{x_1}=(-1.05+0.5i,-1+0.5i,-0.9+0.5i)$, $A^j_{x_2}=(-3.2+1.5j,-3+1.5j,-2.6+1.5j)$, $(i,j=0,1,2,3,4)$. The parameters of the proposed controller are set as $\boldsymbol{H}_{\mathrm{d}}=\mathrm{diag}(1,1)$, $\boldsymbol{D}_{\mathrm{d}}=\mathrm{diag}(10,10)$, $\boldsymbol{K}_{\mathrm{d}}=\mathrm{diag}(8,8)$, $\eta_{\mathrm{m1}}=7.5$, $\eta_{\mathrm{m2}}=8.2$, $\eta_{\mathrm{u1}}=3$, $\eta_{\mathrm{u2}}=6.5$, $c_1=45.8$, $c_2=43.4$, $c_3=44.3$, $c_4=9.2$, $c_5=9.2$, $c_6=10.2$. The arbitrary convergence time of ACTNDO and that of the proposed controller are set as $\Delta t_{\mathrm{o1}}=1.5$s, $\Delta t_{\mathrm{o2}}=1.5$ s and $\Delta T_{\mathrm{u}}=4$ s, $\Delta T_{\mathrm{m}}=4$ s, respectively.

The desired trajectory of the quadrotor and the onboard manipulator are set to $\boldsymbol{q}_{\mathrm{d}}=[1+0.5\sin(\pi/10t),-0.5+0.5\sin(\pi/10t)]^{\mathrm{T}}$ and $\boldsymbol{p}_{\mathrm{d}}=[\sin(\pi/18t),t,2]^{\mathrm{T}}$, respectively. The initial condition is set as $\boldsymbol{q}=[0,0]^{\mathrm{T}}$, $\boldsymbol{p}_{\mathrm{b}}=[0.4,0.4,0.4]^{\mathrm{T}}$, $\boldsymbol{\Phi}_{\mathrm{b}}=[0,0,0]^{\mathrm{T}}$. The interaction forces $\boldsymbol{\tau}_{\mathrm{ext}}=[\tau_{\mathrm{ext1}},\tau_{\mathrm{ext2}}]^{\mathrm{T}}$ are set as $\tau_{\mathrm{ext1}}=2+0.3\sin(\pi/8t)+0.2\sin(\pi/10t)$, $\tau_{\mathrm{ext2}}=2+0.3\sin(\pi/10t)+0.2\sin(\pi/6t)$. The lumped disturbances of the quadrotor are denoted as $\Delta_{\mathrm{u}}=[\Delta_{\mathrm{u1}},\Delta_{\mathrm{u2}},\Delta_{\mathrm{u3}},\Delta_{\mathrm{u4}},\Delta_{\mathrm{u5}},\Delta_{\mathrm{u6}}]^{\mathrm{T}}$ with $\Delta_{\mathrm{u1}}=\Delta_{\mathrm{u2}}=\Delta_{\mathrm{u3}}=0.2\sin(\pi/8t)$ and $\Delta_{\mathrm{u4}}=\Delta_{\mathrm{u5}}=\Delta_{\mathrm{u6}}=0.1\sin(\pi/8t)$.

For interaction force simulations, comparative studies are performed with the external wrench estimator (EWE) as detailed in [23]. The online parameter estimator (OPE) as specified in [24] is adopted to comparatively verify the effective estimation performance of TSKFE. To demonstrate the performance of the arbitrary convergence time controller, comparisons are conducted using the nonsingular terminal sliding mode surface with arctan function (ATNTSM) as studied in [17].

B. Analysis of Results

The comparative results between ACTNDO and EWE regarding the interaction force estimation and their errors are shown in Fig. 2. From Fig. 2(a), it is observed that the precise and effective estimation of unknown interaction forces at arbitrary convergence time $\Delta T_{\mathrm{o}}=3$ s is achieved by ACTNDO both at the start and the end of interaction. The estimated error convergence rate of the proposed approach is faster than that of the comparative method, as illustrated by Fig. 2(b). Moreover, the smaller convergence amplitudes and steady-state errors are

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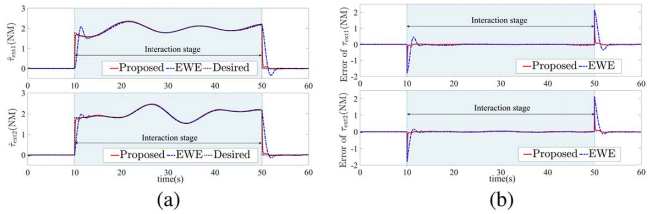


Fig. 2. Simulation-based comparative results of estimation performance and errors for interaction force with ACTNDO and EWE. (a) Estimation of interaction forces acting on two joints of onboard manipulator. (b) Estimation errors of interaction forces acting on two joints of onboard manipulator.

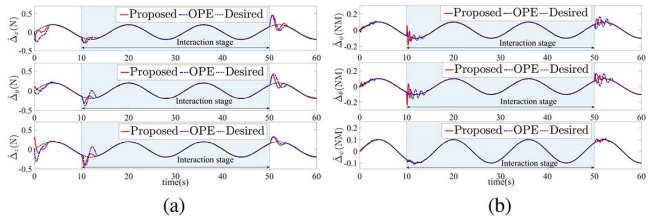


Fig. 3. Simulation-based comparative results of estimation performance for lumped disturbances with TSKFE and OPE. (a) Estimation of position lumped disturbances. (b) Estimation of attitude lumped disturbances.

exhibited by the proposed method. That is, the proposed ACTNDO exhibits excellent estimated performance.

The comparative outcomes of estimating the position and attitude lumped disturbances for quadrotor are depicted in Fig. 3(a) and 3(b), respectively. Despite larger convergence amplitudes for roll and pitch lumped disturbance in terms of estimation errors when compared to the reference method OPE, the response rates and steady-state performances overall position and attitude lumped disturbances by TSKFE demonstrate slight improvements. Interestingly, the number of the newly added fuzzy rules for TSKFE reaches 1295 over the initial sparse fuzzy rule base. That is, TSKFE entails favorable estimation performance and rapid convergence rate, exhibiting the robustness, while helping increase future running efficiency by dynamically improving the knowledge base itself.

The comparative results of tracking performance regarding onboard manipulator and quadrotor, between running the proposed controller and the reference ATNTSM controller with the same ACTNDO and TSKFE, are shown in Fig. 4(a) and 4(b), respectively. The position and attitude tracking errors of quadrotor are depicted in Fig. 4(c) and 4(d), respectively. From these figures, it is observed that the trajectory tracking convergence rates of the proposed controller for both onboard manipulator and quadrotor are faster while having smaller steady-state errors as compared to the reference ATNTSM controller. Particularly, the tracking errors converge to a bounded area with the equilibrium state at the arbitrary convergence time $\Delta T_{\mathrm{m}} = 4\,\mathrm{s}$ and $\Delta T_{\mathrm{u}} = 4\,\mathrm{s}$. The root-mean-square error (RMSE) and the standard deviation (STD) are provided to have a clear and intuitive examination of tracking performances in Table I. Overall, excellent trajectory tracking with high accuracy and rapid response at arbitrary convergence time is demonstrated by the proposed controller.

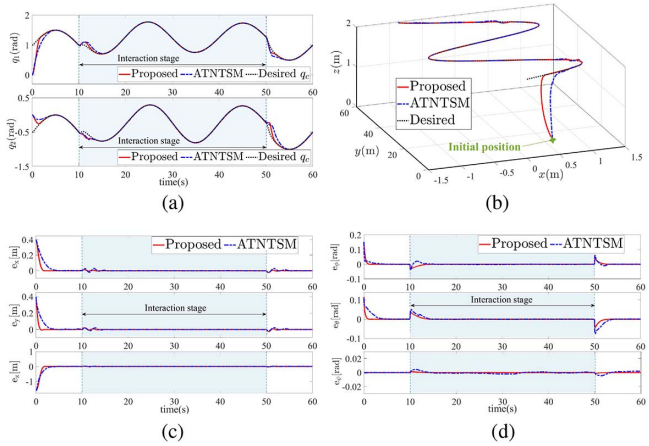


Fig. 4. Simulation-based comparative results of trajectory tracking performance and tracking errors of onboard manipulator and quadrotor, between proposed controller and ATNTSM controller. (a) Trajectory tracking performances of two joints of onboard manipulator. (b) Position trajectory tracking performances for quadrotor in 3-D space. (c) Position tracking errors of quadrotor. (d) Attitude tracking errors of quadrotor.

TABLE I
STEADY-STATE ERROR STATISTICS OF TRAJECTORY TRACKING IN SIMULATIONS

State	Proposed Controller		ATNTSM Controller in [17]	
	RMSE	STD	RMSE	STD
x(m)	0.0156	0.6777	0.0231	0.6799
y(m)	0.0109	1.7016	0.0174	1.7309
z(m)	0.1575	0.1498	0.1618	0.1560
ϕ(rad)	0.0086	0.0355	0.0093	0.0425
θ(rad)	0.0074	0.0422	0.0149	0.0591
ψ(rad)	0.0002	0.0014	0.0014	0.0023
q_1(rad)	0.1036	0.3954	0.1141	0.3956
q_2(rad)	0.0589	0.3801	0.0756	0.3855

V. EXPERIMENTS AND RESULTS ANALYSIS

Further to simulation-based investigations, real-world experimentation has also been carried out, reported as follows.

A. Experimental Setup

The UAM under investigation is constructed for active interaction and dynamic tracking, the hardware configuration and the experimental setup are shown in Fig. 5. A three-joint onboard manipulator is developed for this experimentation, and a 1-D force sensor is equipped on the end effector for real-time measurement of interaction force. The measured force is not used as control feedback but is only utilized to validate the estimated performance of ACTNDO. A combination mobile robot designed with an interaction platform based on rigid carbon plates and preinstalled visual tag is adopted as the dynamics target. The mobile robot is controlled by a human operator and moves in a straight line at a constant speed of $0.1\,\mathrm{m/s}$. The onboard manipulator is maintained in compliant contact with the interaction platform of the dynamic mobile robot. Images captured by a high-frame-rate monocular camera are used for visual localization in the UAM. An onboard computer is utilized for processing data from the end-effector force sensors and

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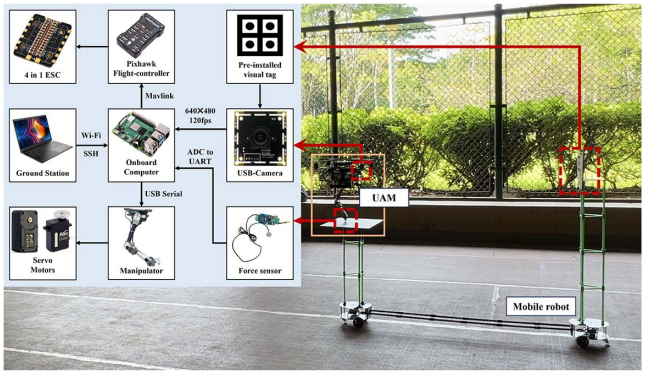


Fig. 5. Experimental setup for realized UAM system.

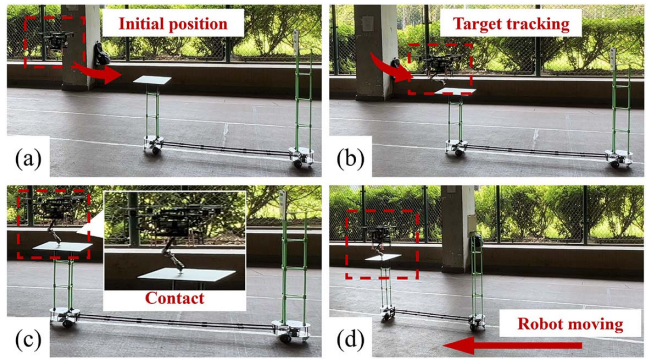


Fig. 6. Snapshots of mobile robot tracking interaction experiment: (a) and (b) process of tracking the mobile robot from the initial position for the UAM; (b) and (c) process of compliant contact of the onboard manipulator with the smooth carbon plate; and (c) and (d) process of dynamic tracking and compliant interaction along with mobile robot for the UAM.

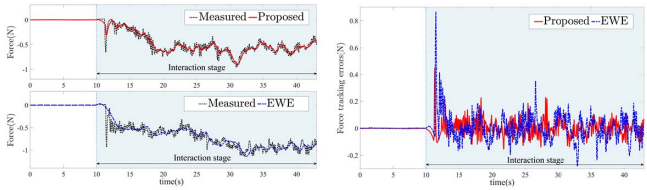


Fig. 7. Comparative experimental results: tracking performance and estimation errors regarding interaction force in vertical z-axis using proposed and compared methods. (a) Estimation of interaction force in vertical z-axis. (b) Estimation errors on interaction force.

executing control strategies. The connection to the flight controller through Mavlink enables the transmission of control signals. Communication with the onboard computer is established by the ground station through wireless fidelity (WIFI) and secure shell (SSH).

The dynamics model parameters of the UAM system are set as $m_u = 4.1\,\text{kg}$ and $g = 9.81\,\text{m/s}^2$. The length of three links fabricated using 3-D printing for onboard manipulator is set as $l_{m1} = 0.126\,\text{m}$, $l_{m2} = 0.092\,\text{m}$, and $l_{m3} = 0.045\,\text{m}$, respectively. $\hat{\boldsymbol{F}}_{ext} = \boldsymbol{J}_m^{-T}\hat{\boldsymbol{\tau}}_{ext}$ is the estimation of interaction force in the Cartesian coordinate system, where $\boldsymbol{J}_m$ is the Jacobian matrix. Triangular membership function is chosen as the membership function. The parameters of ACTNDO are set as $\lambda = 8.2$, $\eta_{o1} = 2.11$, $\eta_{o2} = 2.63$. The parameters of TSKFE are set as $k = 3$, $\mu_{max} = 0$, $A^i_{x_1} = (-1.05 + 0.5i, -1 + 0.5i, -0.9 + 0.5i)$, $A^j_{x_2} = (-3.2 + 1.5j, -3 + 1.5j, -2.6 + 1.5j)$, $(i,j = 0,1,2,3,4)$. The parameters of the proposed controller are set as $\boldsymbol{H}_d = \text{diag}(1,1,1)$, $\boldsymbol{D}_d = \text{diag}(4,4,4)$, $\boldsymbol{K}_d = \text{diag}(9,9,9)$, $\eta_{m1} = 5.3$, $\eta_{m2} = 4.8$, $\eta_{u1} = 2.2$, $\eta_{u2} = 2.75$, $c_1 = 20.9$, $c_2 = 31.1$, $c_3 = 33.3$, $c_4 = 3.1$, $c_5 = 4.22$, $c_6 = 3.75$. The arbitrary convergence time of ACTNDO and proposed controller for the UAM are set as $\Delta T_o = 4\,\text{s}$ and $\Delta T_u = 6\,\text{s}$, $\Delta T_m = 6\,\text{s}$, respectively.

B. Analysis of Results

The snapshots taken over complete tacking interaction processes of the UAM are depicted in Fig. 6, it is observed that the dynamic compliant tracking interaction on the smooth carbon plate for the mobile robot is effectively executed by the UAM. In the first stage of the tracking interaction experiment, the UAM is manually operated to take off to the designated initial position. Afterward, the switch is made to the proposed control strategy, allowing the UAM to smoothly reach above the mobile robot within arbitrary convergence time $\Delta T_u = 6\,\text{s}$, as shown in Fig. 6(a) and 6(b). In the contact stage, the onboard manipulator is controlled by the proposed method to track the desired trajectory, exhibiting compliant characteristics upon contact with the carbon plate and maintaining perpendicular contact, as shown in Fig. 6(b) and 6(c). Finally, in dynamic tracking interaction stage, the proposed control strategy drives the UAM to perform tracking compliant contact according to the mobile robot trajectory, as shown in Fig. 6(c) and 6(d).

The comparative experimental results regarding the interaction force estimation and the errors are shown in Fig. 7. The interaction force is distributed along the z-axis of the UAM body coordinate system according to the onboard three-joint manipulator maintaining perpendicular contact with the carbon plate on the mobile robot. It is measured by a 1-D force sensor to obtain the actual contact force. From the estimation error results in Fig. 7(b), it is observed that the estimation error of the proposed method converges within the steady-state error range at arbitrary convergence time $\Delta T_o = 4\,\text{s}$, with a faster convergence rate and smaller convergence amplitude. Also, the estimation of interaction force for ACTNDO is maintained around the actual force, with the tracking accuracy of interaction force being approximately $\pm 0.2\,\text{N}$, verifying the effectiveness and robustness of the proposed approach.

The experimental results on tracking errors for the three-joints onboard manipulator are shown in Fig. 8(a), comparing between the proposed controller and the ATNTSM controller with ACTNDO. The same proposed method with OPE is adopted as the reference approach to contrasting the influence of TSKFE and OPE on trajectory tracking performances for quadrotor. Specifically, the number of the newly added fuzzy rules in TSKFE reaches 203 at the end of the experiments. The comparative results of position and attitude tracking errors are shown in

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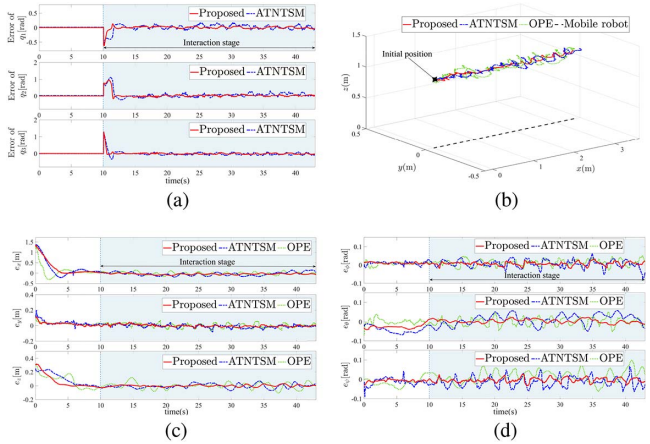


Fig. 8. Comparative experimental results: trajectory tracking performance and tracking errors for UAM implemented with proposed or reference method. (a) Trajectory tracking errors of the three-joint onboard manipulator. (b) 3-D trajectory for the UAM during dynamic tracking interaction. (c) Position trajectory tracking errors for quadrotor. (d) Attitude trajectory tracking errors for quadrotor.

TABLE II
EXPERIMENTAL STEADY-STATE ERROR STATISTICS OF TRAJECTORY TRACKING

State	Proposed Controller		ATNTSM Controller in [17]		Proposed Method With OPE in [24]	
	RMSE	STD	RMSE	STD	RMSE	STD
x(m)	1.0970	0.9197	1.9022	1.0178	1.2970	1.1172
y(m)	0.0347	0.0556	0.0385	0.0598	0.0413	0.0575
z(m)	0.0421	0.0783	0.0471	0.1432	0.0518	0.0796
ϕ(rad)	0.0162	0.0097	0.0257	0.0242	0.0166	0.0148
θ(rad)	0.0460	0.0160	0.0591	0.0164	0.0552	0.0290
ψ(rad)	0.0508	0.0018	0.0618	0.0023	0.0633	0.0047
q_1(rad)	0.0469	0.0199	0.0653	0.0254	/	/
q_2(rad)	0.0182	0.0238	0.0299	0.0601	/	/
q_3(rad)	0.0151	0.0340	0.0170	0.0500	/	/

Fig. 8(c) and 8(d), respectively. The compared results of 3-D trajectory for UAM is shown in Fig. 8(b).

It can be seen from Fig. 8(a) that the trajectory tracking accuracy is approximately ± 0.15 rad during the tracking interaction stage. From Fig. 8(c) and 8(d), it is revealed that the position tracking accuracy and attitude tracking accuracy of interaction are maintained within ± 0.06 m and ± 0.08 rad, respectively. The position trajectory tracking errors of the proposed method are smooth, with small convergence amplitudes. Particularly, the convergence time of onboard manipulator and quadrotor is $\Delta T_{\mathrm{m}} = 6$ s and $\Delta T_{\mathrm{u}} = 6$ s, respectively. The steady-state errors are smaller and the response rate is faster than that of each compared method, verifying the effectiveness and feasibility of the proposed control architecture. To reinforce this observation, the steady-state error statistics of trajectory tracking for the UAM are provided in Table II.

VI. CONCLUSION

In this article, a dynamic tracking interaction control architecture within arbitrary convergence time has been presented to achieve both compliant aerial interaction and stable tracking flight. To implement effective force tracking and compensation of unknown interferences, an ACTNDO and a TSKFE with k-FRI have been designed to accurately estimate the interaction force and lumped disturbances, respectively. An arbitrary convergence time control strategy has been introduced for UAM to guarantee precise tracking control with rapid response. The stability of the proposed control architecture has been analyzed through Lyapunov theory, and its feasibility and performance have been shown through both simulations and real-world investigation by tracking and touching mobile robot dynamically. For future work, the continuous interaction tasks involving complex aerial interaction missions will be extended, such as aerial supply and wind turbine blade contact inspection.

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